

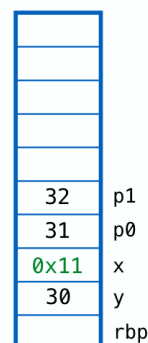
Worksheet 8B

Quiz 1: *Garbage in the middle*

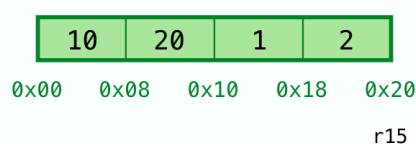
Which cells are garbage?

- a) 0x00 and 0x08
b) 0x08 and 0x10
c) 0x18 and 0x20
d) None
e) All

```
let y = let tmp = (10, 20)
        in tmp[0] + tmp[1]
    , x = (1, 2)
    , p0 = x[0] + y
    , p1 = x[1] + y
in (p0, p1)
```



Lets reclaim & recycle garbage!



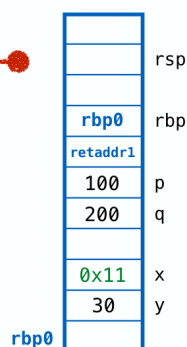
Quiz 2: Garbage in the middle with Stack

Which cells are garbage?

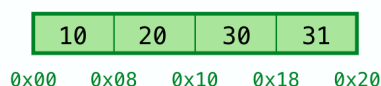
- 0x00 and 0x08
- 0x08 and 0x10
- 0x10 and 0x18
- None
- All

```
def foo(p, q):
    let tmp = (p, q)
    in tmp[0] + tmp[1]

let y = foo(10, 20)
, x = (y, y + 1)
, z = foo(100, 20)
in
    x[0] + z
```



Lets reclaim & recycle garbage!



Quiz 3: *Recursive Data*

What is the heap when `range(0, 3)` returns?

```
def range(i, j):
    if (j <= i): false else: (i, range(i+1, j))

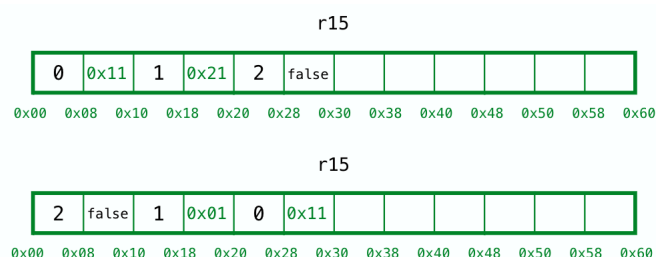
def sum(l):
    if l == false: 0 else: l[0] + sum(l[1])

let t1 =
    let l1 = range(0, 3)
    in sum(l1)
, l = range(t1, t1 + 3)
in
    (1000, l)
```



(A)

(B)



Quiz 4: Recursive Data

```
def range(i, j):
    if (j <= i): false else: (i, range(i+1, j))

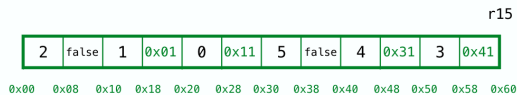
def sum(l):
    if l == false: 0 else: l[0] + sum(l[1])

let t1 =
    let l1 = range(0, 3)
    in sum(l1)
, l = range(t1, t1 + 3)
in
(1000, l)
```

call range(3,6)

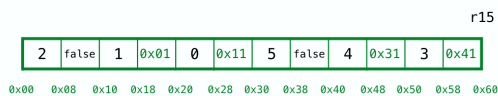
What is the value of `l` at the point indicated by the red arrow?

- a) 0x30
- b) 0x31
- c) 0x50
- d) 0x51
- e) 0x60

**Quiz 5: Live Cells**

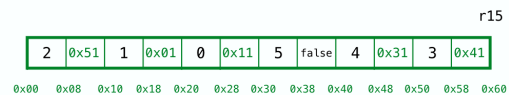
QUIZ: Which cells are “live” on the heap?

- (A) 0x00
- (B) 0x10
- (C) 0x20
- (D) 0x30
- (E) 0x40
- (F) 0x50



QUIZ: Which cells are “live” on the heap?

- (A) 0x00
- (B) 0x10
- (C) 0x20
- (D) 0x30
- (E) 0x40
- (F) 0x50

**Quiz 6: What should be printed?**QUIZ: What should `print(0x21)` show?

- (A) (0, (1, (2, false)))
- (B) (3, (4, (5, false)))
- (C) (0, (1, (2, (3, (4, (5, false))))))
- (D) (3, (4, (5, (0, (1, (2, false))))))
- (E) (2, (1, (0, (3, (4, (5, false))))))

**Quiz 7: Your turn!**

What is something you found confusing in today’s lecture (or earlier)?